Signal iOS 8.36 (build 1981)

Source-to-binary verification of the App Store build

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Result. The compiled code (`__TEXT,__text`) of all 21 binaries in the App Store app is **byte-identical** to a local build of tag `8.36.0.1981` with Xcode 27.1 (27A9275). That covers Signal's five own binaries and all 16 bundled frameworks. Every other difference is one already explained for earlier versions: the signature-size field in every binary; the UUID and the linker's GOT-slot order in `SignalServiceKit` alone, with every stub resolving to the same target; build metadata in plists; and the app icons, which are rasterised at build time and differ by at most 1/255.

Its four prebuilt native libraries (libsignal, RingRTC, WebRTC, libmobilecoin) were also rebuilt from their own public source and match, so 8.36 is **fully reproducible from public source**. That covers not just the Signal-iOS repository but every library it bundles.

1 Method

This is the same method as for 8.35.1, which describes it in full.

1. The source is cloned at the Xcode Cloud path `/Volumes/workspace/repository`.
2. `make dependencies` fetches the dependencies.
3. An unsigned `xcodebuild archive` runs with the scheme's App Store Release configuration, using the exact Xcode build recorded in the shipped `Info.plist`.
4. `verify/compare_build.sh` and `verify/explain_residuals.sh` compare the result with the decrypted App Store IPA.

Input	Value
App Store app	Signal 8.36 (1981), downloaded from the App Store (version ID `894738344`) and decrypted with unfairplay. The encrypted IPA passes all 26,778 of its code-signature page hashes. Decryption changed nothing outside the FairPlay-encrypted ranges apart from the `cryptid` flag.
Source	Tag `8.36.0.1981`, commit `012f2f2`, the commit recorded in the shipped `Info.plist`. Pods submodule `fae00ed`; libsignal `0.106.1` (Swift Package Manager) and RingRTC `v2.73.0`, at the versions the tag pins.
Toolchain	Xcode 27.1 (27A9275), matching the shipped `DTXcodeBuild`.

2 Results

Check	Result
`__TEXT,__text`, all 21 binaries	Byte-identical (68.2 MB, hashed with Apple's `segedit` and `shasum`)
Functions, Signal's 5 binaries	150,344 of 150,344 match after address normalisation
Strings, symbols, linked libraries	Identical
Header bytes	`__LINKEDIT` size (signature) in all 21 binaries; `LC_UUID` in `SignalServiceKit` only

Check	Result
GOT slot order	`SignalServiceKit` only (999 + 1 stub bytes). All 1,873 + 999 stubs resolve to the same target.
Plists	38 differ, only in `BuildDetails` time, `BuildMachineOSBuild`, `CFBundleVersion` and `DTAppStoreToolsBuild`
Asset catalogs	All renditions identical (2,832 main, 2,542 share extension, 8 NSE), except the 72 rasterised app-icon bitmaps, which `actool` renames randomly and which differ slightly in content
Loose icons	107 of 14,400 and 176 of 23,104 pixels differ, by at most 1/255 in one channel
Prebuilt native libraries	All four rebuilt from their own public source and matched: libsignal `v0.106.1` byte-identical (803 of 803 members, SHA-256 `b7f79b9f…`, the device slice of the pinned XCFramework `60baf6a3…`); RingRTC `v2.73.0`, WebRTC `8037a` and libmobilecoin `6.0.2` are the same builds verified for 8.35.1 (byte-identical, WebRTC apart from its 8-byte UUID). See below.

3 Prebuilt native libraries

libsignal `v0.106.1` was built from tag `v0.106.1` exactly as Signal's public CI run of 27 October 2026 did: `swift/build_ffi.sh --release` for `aarch64-apple-ios`, with Rust 1.98.1 and Xcode 16.4 at the runner's paths. That CI run's log shows the pinned checksum `60baf6a3…`. The library is byte-identical to the `ios-arm64` slice of that XCFramework, which 8.36 now fetches through Swift Package Manager. The other three prebuilt libraries are unchanged from 8.35.1 and were rebuilt for that version (8.35.1 report, section "Prebuilt native libraries rebuilt from source"). libmobilecoin now arrives as `libmobilecoin-ios-build-v6.0.2.zip` from `build-artifacts.signal.org` instead of a git repository, but its `libmobilecoin.a` is the identical file (`ebfa6710…`). With the app's own code also byte-identical, every compiled byte in 8.36 traces back to public source: the equivalent of a source-reproducible build. Scripts: `verify/deps/`; outputs: `out/deps/`.

4 Limitations

As for earlier versions:

- The public source itself is not reviewed.
- The toolchains are trusted: Apple's Xcode compilers, the Rust toolchains from rustup and Chromium's prebuilt clang. A compromised compiler would reproduce its own output.
- The causes of the UUID and icon-rasterisation differences are inferred, not proven.

5 Reproducing

```
git clone https://github.com/signalapp/Signal-iOS.git /Volumes/workspace/repository
cd /Volumes/workspace/repository && git checkout 8.36.0.1981 && make dependencies
# xcodebuild archive with Xcode 27.1 (27A9275), as in the 8.35.1 report, then:
verify/compare_build.sh signal-ios-8.36-1981.decrypted.ipa <built Signal.app>
versions/8.36.0.1981/out/ipa_vs_build
verify/explain_residuals.sh signal-ios-8.36-1981.decrypted.ipa <built Signal.app>
versions/8.36.0.1981/out/residuals
```

The decrypted IPA is attached to the `signal-8.36` release. The app is also in `versions/8.36.0.1981/signal_dec/`, and all outputs are in `versions/8.36.0.1981/out/`.

6 Changelog

Date	Change
8 November 2026	First published version.
8 November 2026	**Added** the prebuilt native libraries: all four rebuilt from source and matched (libsignal `v0.106.1` here; the rest shared with 8.35.1). **Changed** the limitation that said they were only checked against their pins. **Added** this changelog.